An $\widetilde{O}(n^{1.5} + n\varepsilon^{-2})$ FPRAS for #Knapsack

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Abstract

This memo is written to be read alongside Feng and Jin's $\widetilde{O}(n^{1.5}\varepsilon^{-2})$ FPRAS for #Knapsack [1]; we use their notation and point into their sections throughout. The claim: one change to their pipeline – refining the rounding grid by a factor n/ℓ, which their sampler (Lemma 4.7) cannot afford but a level-enumerating sampler gets for free – drops the total running time to

$$\widetilde{O}\big(n^{1.5} + n\,\varepsilon^{-2}\big)\,,$$

uniformly in ε and in their popular-class parameter ℓ. The ε-dependence then equals the number of samples the estimator draws, times the cost of writing one down.[1]

1 Their pipeline, and where its time goes

Recall the structure of [1]. The count is approximated as $|\Omega'| \cdot p$: here Ω' is the solution set of a rounded instance, counted exactly by their merge tree of count arrays (§4–§5), and $p = |\Omega|/|\Omega'|$ is estimated by drawing uniform samples from Ω' (§5.3) and checking true feasibility. Their time is

$$\underbrace{\widetilde{O}(n^{1.5})}_{\text{construction (Thm 4.2/6.1, L.5.9)}} \quad + \quad \underbrace{N \cdot \widetilde{O}(\ell\sqrt{n})}_{N \text{ queries of Lemma 4.7}}\quad , \qquad N = \widetilde{O}\Big(\frac{n}{\ell\,\varepsilon^2}\Big) \quad \text{(from Lemma 3.3)},$$

with $\ell \in [2, 8n]$ the popular-class parameter of Lemma 3.1. The two ℓ's cancel in the second term: $N \cdot \ell\sqrt{n} = n^{1.5}\varepsilon^{-2}$ for every ℓ – which is why the paper is free to choose ℓ for the benefit of the first term.

The size of N traces back to the grid. Their rounding is coarse enough that the rounded superset overshoots: $|\Omega'|/|\Omega| = \widetilde{O}(n/\ell)$, so p can be as small as ℓ/n, and estimating a ratio that small to relative ε costs $1/(p\varepsilon^2)$ samples – exactly their N. The overshoot is rounding debt: a mask's rounded weight is below its true weight by up to the accumulated rounding error, so masks whose true weight lands in that error band above T are counted as solutions although they are not. Lemma 3.3 is precisely the statement that a band of width T/ℓ carries a polylogarithmic multiple of $|\Omega|$; their error is n/ℓ bands wide, hence the n/ℓ overshoot.

2 The observation

Nothing in the correctness of the pipeline forces the coarse grid. What forces it is the query of Lemma 4.7: it reads array segments, so its cost scales with array length, and refining the grid

[1] A Lean 4 development accompanying this memo machine-checks the components introduced here, including the overcount bound of Section 3; see the repository.

by a factor F multiplies every array length – and with it the whole sampling phase – by F. The coupling between grid and ℓ is an artifact of *how they sample*.

In earlier work on this repository we replaced the Lemma-4.7 query by a pruned witness-split sampler: it descends the merge tree splitting each position among the *witness levels* of the two children (the levels of their floating-point representation, §6), and its per-draw work enumerates levels – a quantity bounded by the subtree's item count – never array positions. Its per-sample cost is $\widetilde{O}(n)$, independent of the grid. That frees the grid:

> *Make the grid fine enough that the rounded count is honest, and pay for it only in array length – which the sampler never reads.*

3 What the finer grid buys

Refine every node's grid by $F = n/\ell$, so the total round-down error of any mask stays below a single band width T/ℓ. Then:

1. **The overshoot collapses to polylog.** Rounding down never loses a solution, so $\Omega \subseteq \Omega'$; and every over-counted mask has true weight in $(T, T + T/\ell]$ – one band. Lemma 3.3 bounds that band by a polylogarithmic multiple of $|\Omega|$, and it does so as a statement about *true* weights: it is blind to the rounding and applies verbatim. Hence $p \geq 1/\text{polylog}$ and $N = \widetilde{O}(\varepsilon^{-2})$.

2. **Construction grows only linearly in F.** A merge costs length $\times \sqrt{\text{levels}}$ (Thm 6.1 is parametric in the array length; the level count is a property of the counts, not the grid). Summed over the class tree this is $\widetilde{O}(F\ell\sqrt{n})$, which at $F = n/\ell$ is $\widetilde{O}(n^{1.5})$ – the ℓ cancels here too, now in our favor.

4 The algorithm

1. Run the reduction, class decomposition (Lemma 3.1), and two-phase structure (§5.4) of [1] unchanged, with every node's grid step a factor n/ℓ finer.

2. Build the merge tree with their Theorem 6.1 subroutine at the finer grid. Alongside each node, store static per-(level, residual-class) sorted position arrays with prefix counts.

3. Draw $N = \widetilde{O}(\varepsilon^{-2})$ samples with the pruned witness-split sampler; every mass and rank query during a descent is a prefix lookup on the static arrays.

4. Check each sampled mask's true weight; output $|\Omega'|$ times the observed feasible fraction, median-amplified.

5 Why this is $n^{1.5} + n\varepsilon^{-2}$

Construction: length $\times \sqrt{\text{levels}}$ per merge, at boosted length $F\ell/2^{h/2}$ and $n_m/2^h$ levels, sums over depths and classes to $F\ell\sqrt{n} \cdot \text{polylog} = n^{1.5} \cdot \text{polylog}$ at $F = n/\ell$. Sampling: $\widetilde{O}(\varepsilon^{-2})$ samples at $\widetilde{O}(n)$ each – level enumeration, grid-independent – is $\widetilde{O}(n\varepsilon^{-2})$. There is no third term.

6 Discussion

What is verified. The statistical heart is machine-checked in Lean: the sandwich $\Omega \subseteq \Omega' \subseteq \Omega \cup \text{band}$ and the overcount bound $100\,|\Omega'| \le (101 + 200h)\,|\Omega|$, proved by composing with our formalization of their Lemma 3.3; likewise the sampler's exactness and the cost ledgers.

What is inherited from [1]. Theorem 6.1 as published, applied at longer arrays (its statement is parametric in length); the two-phase pipeline of §5.4; and Lemma 3.3's instance-side hypotheses, discharged by their own reduction.

What remains. We prove no lower bound, and claim no optimality. At $\varepsilon = \Theta(1)$ the bound is $\widetilde{O}(n^{1.5})$, tying [1]: the construction itself, whose improvement is the open problem they pose in §1.3 (tied to the bounded-monotone $(\max, +)$-convolution frontier). That is now the only ε-regime where the time exceeds the sample size: for every ε, sampling costs $\widetilde{O}(n\varepsilon^{-2})$ – ε^{-2} samples at the cost of writing one down. The price paid is space: the boosted arrays occupy $\widetilde{O}(n^{1.5})$ words, the same order as their construction time but more than their unboosted arrays.

References

[1] W. Feng, C. Jin. *A Faster Algorithm for Counting Knapsack Solutions*. SODA 2025. arXiv:2410.22267.

[2] M. Dyer. *Approximate counting by dynamic programming*. STOC 2003.

[3] P. Gawrychowski, L. Markin, O. Weimann. *A Faster FPTAS for #Knapsack*. ICALP 2018.